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SHEET FEEDING DEVICE AND IMAGE FORMING APPARATUS

Abstract

A sheet feeding device includes: a sheet housing unit that is mounted on a body of an apparatus such that the sheet housing unit can be pulled out of the body of the apparatus; a blowing unit that blows air toward a sheet housed in the sheet housing unit; a heating unit that heats the air blown by the blowing unit; and a control unit that controls the blowing unit and the heating unit. In the sheet feeding device, the control unit is configured to drive the blowing unit and the heating unit during a sheet feeding operation of feeding the sheet from the sheet housing unit, and continuously drive the blowing unit and stop the heating unit when the sheet housing unit is pulled out of the body of the apparatus during the sheet feeding operation.

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Background/Summary

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2024-029749 filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a sheet feeding device and an image forming apparatus including the same.

BACKGROUND

[0003] An electrophotographic image forming apparatus includes a sheet feeding device that feeds a sheet to an image forming unit. The sheet feeding device takes one sheet from a plurality of sheets stacked in a tray and feeds the taken sheet to the image forming unit. As the sheets housed in the tray, not only sheets of paper referred to as so-called plain paper but also special sheets such as so-called coated paper whose surface has been subjected to coating treatment, and the like are known. In the past, it has been difficult to separate the sheets stacked in the tray in some cases because they have been in close contact with each other, resulting in double feeding. The sheets are likely to come into close contact with each other in high humidity conditions.

[0004] In order to prevent the sheets from coming into close contact with each other, blowing warm air onto the sheets stacked in the tray has been known in the past. For example, a sheet feeder that does not blow warm air at excessive temperature onto the sheets when replacing them has been proposed.

SUMMARY

[0005] A sheet feeding device according to an aspect of the present disclosure includes: a sheet housing unit that is mounted on a body of an apparatus such that the sheet housing unit can be pulled out of the body of the apparatus; a blowing unit that blows air toward a sheet housed in the sheet housing unit; a heating unit that heats the air blown by the blowing unit; and a control unit that controls the blowing unit and the heating unit. In the sheet feeding device, the control unit is configured to drive the blowing unit and the heating unit during a sheet feeding operation of feeding the sheet from the sheet housing unit, and continuously drive the blowing unit and stop the heating unit when the sheet housing unit is pulled out of the body of the apparatus during the sheet feeding operation.

[0006] An image forming apparatus according to another aspect of the present disclosure includes: the sheet feeding device; and an image forming unit that forms an image on the sheet fed by the sheet feeding device.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF FIGURES

[0008] FIG. 1 is a perspective view showing a configuration of an image forming apparatus according to an embodiment of the present disclosure.

[0009] FIG. 2 is a cross-sectional view showing an internal configuration of the image forming

apparatus according to the embodiment of the present disclosure.

[0010] FIG. 3 is a cross-sectional view showing a configuration of a sheet housing unit included in the image forming apparatus according to the embodiment of the present disclosure.

[0011] FIG. 4 is a block diagram showing a configuration of a control unit of the image forming apparatus.

[0012] FIG. 5 is a flowchart showing a first processing example of the procedure of sheet blowing processing executed by the control unit of the image forming apparatus.

[0013] FIG. 6 is a flowchart showing a second processing example of the procedure of the sheet blowing processing executed by the control unit of the image forming apparatus.

[0014] FIG. 7 is a flowchart showing a third processing example of the procedure of the sheet blowing processing executed by the control unit of the image forming apparatus.

DETAILED DESCRIPTION

[0015] Embodiments of the present disclosure will be described below with reference to the drawings. Note that the following embodiments are merely examples embodying the present disclosure and do not limit the technical scope of the present disclosure.

[0016] For convenience of description, the vertical direction in the installation state (shown in FIG. 1) in which an image forming apparatus 10 can be used is defined as an up-and-down direction 7. Further, a front-and-rear direction 8 is defined with the surface where a paper feed device 24 (sheet housing device) shown in FIG. 1 is inserted and removed in the installation state as the front (front surface). Further, a right-and-left direction 9 is defined on the basis of the front surface of the image forming apparatus 10 in the installation state.

[0017] The image forming apparatus 10 according to the embodiment of the present disclosure is an apparatus having at least a printing function. The image forming apparatus 10 is a color printer of a so-called tandem type. Note that the image forming apparatus 10 may be a monochrome printer.

[0018] As shown in FIG. 1 and FIG. 2, the image forming apparatus 10 includes a casing 10A (an example of the body of the apparatus according to the present disclosure). The casing 10A has a substantially rectangular parallelepiped shape as a whole. The respective units constituting the image forming apparatus 10 are disposed inside the casing 10A.

[0019] As shown in FIG. 2, the image forming apparatus 10 includes four image formation units 4, an intermediate transfer belt 5, a light scanning device 13, a secondary transfer roller 20, a fixing device 16, a sheet tray 18, the paper feed device 24, an operation display unit 25 (see FIG. 1), a belt cleaning device 6, a blower fan 35 (an example of the blowing unit according to the present disclosure), a heater 36 (an example of the heating unit according to the present disclosure), a control unit 90 (see FIG. 4), and the like. Note that the image formation unit 4, the intermediate transfer belt 5, the fixing device 16, and the secondary transfer roller 20 realize the image forming unit according to the present disclosure. Further, the image forming apparatus 10 including the paper feed device 24, the blower fan 35, and the heater 36 is an example of the sheet feeding device according to the present disclosure. Note that the sheet feeding device according to the present disclosure only needs to include at least the paper feed device 24, the blower fan 35, and the heater 36.

[0020] Each of the image formation units 4 includes a photoreceptor drum, a charging device, a development device, a primary transfer roller, and the like, and forms an image on the intermediate transfer belt 5 in accordance with an electrophotographic method. The image formation units 4 are arranged along the front-and-rear direction 8 inside the casing 10A and form color images on the basis of a so-called tandem method. The image formation units 4 transfer toner images of respective colors to the intermediate transfer belt 5 in a superimposed manner to form a color toner image on the intermediate transfer belt 5. The toner image on the intermediate transfer belt 5 is transferred to printing paper (an example of the sheet according to the present disclosure) fed from the paper feed device 24, by the secondary transfer roller 20. Note that the image formation unit 4

does not necessarily need to perform image forming processing by an electrophotographic method and may perform image forming processing by, for example, an inkjet recording method.

[0021] The intermediate transfer belt **5** is provided above the four image formation units **4**. The intermediate transfer belt **5** is supported to be driven to rotate by a drive roller **5A** and a driven roller **5B**. By being supported by the drive roller **5A** and the driven roller **5B**, the intermediate transfer belt **5** is capable of moving in the direction indicated by an arrow **19** while the surface of the intermediate transfer belt **5** is in contact with the surface of each photoreceptor drum.

[0022] The light scanning device **13** is provided below the four image formation units **4**. The light scanning device **13** applies laser light to each of the photoreceptor drums of the image formation units **4** on the basis of image data of each color input to form an electrostatic latent image on each photoreceptor drum.

[0023] The paper feed device **24** is provided at the bottom of the casing **10A**. The paper feed device **24** includes a sheet housing unit **41** and a feeding mechanism **32**. A plurality of sheets of printing paper (corresponding to the sheet according to the present disclosure) is housed in the sheet housing unit **41**. A vertical conveying path **26** that extends from the paper feed device **24** to the fixing device **16** through the secondary transfer roller **20** is formed on the rear side of the casing **10A**. The feeding mechanism **32** takes the printing paper stacked in the sheet housing unit **41** one sheet at a time and feeds the taken printing paper to the vertical conveying path **26**.

[0024] The secondary transfer roller **20** is provided at a position facing the drive roller **5A** on the rear side of the casing **10A**. The secondary transfer roller **20** transfers a toner image to the printing paper from the intermediate transfer belt **5**.

[0025] The fixing device **16** is provided above the secondary transfer roller **20**. The fixing device **16** applies heat to printing paper to fix a toner image to the printing paper. The fixing device **16** includes a heating roller **16A** and a pressure roller **16B**, which are a pair of rollers. The heating roller **16A** is disposed on the inner side than the pressure roller **16B**, in the casing **10A**. The heating roller **16A** is heated by a heating device. The printing paper conveyed from the secondary transfer roller **20** to the fixing device **16** is conveyed while being sandwiched between the heating roller **16A** and the pressure roller **16B**. At this time, heat is transferred from the heating roller **16A** to the printing paper, and the toner on the printing paper melts to be fixed to the printing paper. This forms a color image on the printing paper.

[0026] An output path **28** is formed between the fixing device **16** to a sheet output port **27**. The printing paper that has passed through the fixing device **16** is conveyed through the output path **28** by an output roller **23** and is output from the sheet output port **27** to the sheet tray **18**.

[0027] FIG. **3** is a diagram showing the cross section of the sheet housing unit **41**. The sheet housing unit **41** stacks a plurality of sheets of printing paper to be fed to the rear when forming an image. As shown in FIG. **3**, the sheet housing unit **41** has an opening **411** on the upper surface thereof and is formed in a rectangular box shape or tray shape, which has a space **412** capable of housing printing paper therein. The sheet housing unit **41** houses a plurality of sheets of printing paper.

[0028] A lift plate **45** that is lifted in the up-and-down direction **7** is attached to a bottom portion **46** of the sheet housing unit **41**. The lift plate **45** is provided on the rear side of the bottom portion **46**. Printing paper is stacked on the lift plate **45**. The front end of the lift plate **45** is rotatably supported by the bottom portion **46** of the paper feed device **24**, and the rear end of the lift plate **45** is a free end. The lift plate **45** moves in the up-and-down direction **7** by receiving driving force from a motor (not shown). As a result, the lift plate **45** is lifted in the up-and-down direction **7** together with the printing paper.

[0029] The sheet housing unit **41** is mounted on the casing **10A** such that it can be pulled out of the casing **10A**. The sheet housing unit **41** is supported on the casing **10A** such that it can be inserted and removed. The casing **10A** supports the sheet housing unit **41** in a movable manner between a mounting position where it is mounted on the casing **10A** and a removal position where it is

removed from the mounting position.

[0030] A position detection sensor **37** (see FIG. **4**) for detecting whether or not the sheet housing unit **41** is located at the mounting position is provided in the casing **10A**. The position detection sensor **37** is connected to the control unit **90**. In the case where the sheet housing unit **41** is located at the mounting position, the position detection sensor **37** transmits a signal indicating that the sheet housing unit **41** is mounted on the casing **10A** to the control unit **90**. The control unit **90** determines, on the basis of this signal, that the sheet housing unit **41** is located at the mounting position, i.e., the sheet housing unit **41** is mounted on the casing **10A**. Meanwhile, in the case where the sheet housing unit **41** is not located at the mounting position, the position detection sensor **37** transmits a signal indicating that the sheet housing unit **41** is not mounted on the casing **10A** to the control unit **90**. The control unit **90** determines, on the basis of this signal, that the sheet housing unit **41** is not located at the mounting position, i.e., the sheet housing unit **41** is pulled out of the casing **10A**.

[0031] The blower fan **35** is a blower that blows air toward and onto the sheets stacked in the sheet housing unit **41**. As shown in FIG. **3**, the blower fan **35** is provided above the sheet housing unit **41** of the paper feed device **24**. The blower fan **35** is located at a position closer to the rear above the sheet housing unit **41**. The blower fan **35** is mounted on the internal frame such as the casing **10A** such that the air outlet thereof faces downward. Further, the suction port of the blower fan **35** faces upward. For this reason, when the blower fan **35** is driven, it sucks in air from above and blows the air toward the upper surface of the sheets stacked in the sheet housing unit **41**. The blower fan **35** is driven and controlled by the control unit **90**.

[0032] The heater **36** is a heating device for heating the sheets stacked in the sheet housing unit **41**. The heater **36** is provided above the blower fan **35**. The heater **36** heats the air blown by the blower fan **35**. The heater **36** radiates heat downward. The air heated by the heater **36** is sucked in by the blower fan **35** and blown toward the sheets housed in the sheet housing unit **41**. The heater **36** is, for example, a halogen heater or an IH heater. The heater **36** is driven and controlled by the control unit **90**.

[0033] The heater **36** is provided with a temperature sensor **361** such as a thermostat and a thermocut for detecting abnormal temperatures of the heater **36**. The temperature sensor **361** is specifically a thermocut as a protective circuit for protecting the heater **36** from abnormal temperatures and is connected in series with the power supply line of the heater **36**. When the temperature sensor **361** detects an abnormal temperature, the power supply line is cut off to stop the power supply to the heater **36**.

[0034] Incidentally, in the image forming apparatus **10** including the blower fan **35** and the heater **36**, if the blower fan **35** is stopped and the heater **36** continues to heat when the sheet housing unit **41** is pulled out of the casing **10A**, the following problem can arise. For example, if the entire sheet housing unit **41** is not fully pulled out, the printing paper housed in the sheet housing unit **41** is heated in a windless state. In this case, the printing paper becomes hot partially. If the sheet housing unit **41** is returned to the casing **10A** as it is, there is a possibility that the temperature distribution of the printing paper becomes uneven, which has a negative effect on the image quality in the case where the printing paper is fed and an image is formed on the printing paper. Further, even in the case where printing paper is further added to the sheet housing unit **41**, there is a possibility that the heat is transferred to the added printing paper, which deteriorates the image quality in the case where an image is formed on the printing paper. Further, even if no printing paper is housed in the sheet housing unit **41**, the sheet housing unit **41** becomes hot partially in the case where the sheet housing unit **41** is heated in a windless state. In this case, there is a possibility that the heat of the sheet housing unit **41** is transferred to printing paper housed thereafter in the sheet housing unit **41**, which has a negative effect on the image quality in the case where the printing paper is fed and an image is formed on the printing paper. Further, in the case where the heater **36** continues to heat in a windless state, there is a possibility that the temperature in the casing **10A** rises suddenly, which

causes the protective circuit of the temperature sensor **361** of the heater **36** to activate to cut off the power supply line of the heater **36** to stop abnormally.

[0035] On the other hand, in this embodiment, since the control unit **90** performs sheet blowing processing described below (see FIG. 5 to FIG. 7), it is possible to prevent the heater **36** from being abnormally overheated and the sheets of printing paper in the sheet housing unit **41** from coming into close contact with each other even in the case where the sheet housing unit **41** is pulled out of the casing **10A** during a feeding operation of printing paper.

Control Unit **90**

[0036] The control unit **90** controls the image forming apparatus **10** and drives and controls the blower fan **35** and the heater **36** to execute the sheet blowing processing described below.

[0037] As shown in FIG. 4, the control unit **90** includes a CPU **91**, a RAM **92**, a ROM **93**, a storage unit **94**, and the like. The control unit **90** is electrically connected to the blower fan **35**, the heater **36**, the position detection sensor **37**, and the like via a signal line or the like. The blower fan **35** and the heater **36** are driven and controlled by receiving individual control signals from the control unit **90**.

[0038] The CPU **91** is a processor that executes a computer program to execute various types of data processing and predetermined control. The RAM **92** is a computer-readable volatile or non-volatile storage device. The RAM **92** temporarily stores the computer program to be executed by the CPU **91**, data output or referred to by the CPU **91** when executing various types of processing, and the like. The ROM **93** is a non-volatile storage device that stores, in advance, a control program such as a BIOS and an OS for causing the CPU **91** to execute various types of arithmetic processing. The storage unit **94** is a flash memory that stores various types of information. The storage unit **94** stores the control program for executing various types of processing by the control unit **90**, and data, a threshold value, a reference value, and the like to be used for various types of processing. Note that the storage unit **94** may be a non-volatile storage device such as an HDD and an SSD connected to the control unit **90**.

[0039] When the CPU **91** executes various control programs stored in the ROM **93** or the storage unit **94** in advance, the control unit **90** drives and controls the blower fan **35** and the heater **36** to execute the sheet blowing processing described below (see FIG. 5 to FIG. 7).

[0040] As shown in FIG. 4, in order to execute the sheet blowing processing, the control unit **90** includes various processing units such as a heating control unit **95** and an airflow rate control unit **96**. Note that in this embodiment, not all of these processing units are necessarily essential and some of them can be omitted in some cases.

[0041] When the CPU **91** executes various types of arithmetic processing according to the control program, the control unit **90** functions as the various processing units. The control unit **90** or the CPU **91** is an example of the computer or processor that executes the control program. Note that some or all of the processing units included in the control unit **90** may include electronic circuits. Further, the control program may be a program for causing a plurality of processors to function as the various processing units.

[0042] The heating control unit **95** controls the driving or stopping of the heater **36**. Further, the heating control unit **95** controls the output of the heater **36** by feedback control based on a comparison between the detection temperature of a temperature sensor (not shown) that detects the temperature around the sheet housing unit **41** and a predetermined target temperature.

[0043] In this embodiment, the heating control unit **95** drives the heater **36** and controls the heating by the heater **36** during a sheet feeding operation of feeding printing paper from the sheet housing unit **41** when a printing job is input in the image forming apparatus **10**. Further, the heating control unit **95** stops the heater **36** when it is determined that the sheet housing unit **41** has been pulled out of the casing **10A** during the sheet feeding operation.

[0044] The airflow rate control unit **96** controls the driving or stopping of the blower fan **35**. Further, the airflow rate control unit **96** controls the airflow rate when the blower fan **35** blows air.

[0045] In this embodiment, the airflow rate control unit **96** drives the blower fan **35** and controls the air blowing by the blower fan **35** during the sheet feeding operation of feeding printing paper from the sheet housing unit **41** when a printing job is input in the image forming apparatus **10**. Further, the airflow rate control unit **96** does not immediately stop the blower fan **35** but continues to drive the blower fan **35** when it is determined that the sheet housing unit **41** has been pulled out of the casing **10A** during the sheet feeding operation.

[0046] Specifically, the airflow rate control unit **96** reduces the airflow rate of the blower fan **35** from a predetermined reference airflow rate to an airflow rate lower than the reference airflow rate in the case where the sheet housing unit **41** is pulled out of the casing **10A** during the sheet feeding operation.

[0047] Further, the airflow rate control unit **96** counts the elapsed time after the heater **36** is stopped by the heating control unit **95**, and stops the blower fan **35** in the case where it is determined that a predetermined set time has elapsed.

[0048] Specifically, the airflow rate control unit **96** changes the set time in accordance with the drive time during which the heater **36** is driven from the time when driving of the heater **36** started to the time when the sheet housing unit **41** was pulled out of the casing **10A** during the sheet feeding operation.

[0049] For example, in the case where the sheet housing unit **41** is pulled out during the sheet feeding operation, the longer the drive time up to this point, the higher the temperature of the heated heater **36**. For this reason, in this case, the longer the drive time, the airflow rate control unit **96** sets the set time to a longer time. Specifically, in the case where the drive time is longer than a predetermined reference drive time, the airflow rate control unit **96** sets, as the set time, the calculation value obtained by adding an increased value proportional to the time difference between the drive time and the reference drive time to the reference drive time. This allows the residual heat of the heater **36** to be reliably reduced and the heater **36** to be reliably cooled, while the sheet housing unit **41** is pulled out.

[0050] Further, in the case where the sheet housing unit **41** is pulled out during the sheet feeding operation, the shorter the drive time up to this point, the lower the risen temperature of the heater **36**. For this reason, in this case, the shorter the drive time, the airflow rate control unit **96** sets the set time to a shorter time. Specifically, in the case where the drive time is shorter than the reference drive time, the airflow rate control unit **96** sets, as the set time, the calculation value obtained by subtracting a subtraction value proportional to the time difference between the drive time and the reference drive time from the reference drive time. This allows the residual heat of the heater **36** to be efficiently reduced and the heater **36** to be reliably cooled, while the sheet housing unit **41** is pulled out.

[0051] In another embodiment, the airflow rate control unit **96** may set, in the case where the drive time is equal to or longer than a predetermined first threshold value, the set time to a first time longer than a predetermined initial set time. In this case, the airflow rate control unit **96** maintains the set time at the initial set time in the case where the drive time is less than the first threshold value and is equal to or more than a second threshold value that is smaller than the first threshold value. Further, the airflow rate control unit **96** sets, in the case where the drive time is less than the second threshold value, the set time to a second time shorter than the initial set time. Note that the first threshold value and the second threshold value can be set arbitrarily.

[0052] Further, the airflow rate control unit **96** changes, in accordance with the drive time of the blower fan **35** and the heater **36** during the sheet feeding operation, the airflow rate of the blower fan **35** after the heater **36** is stopped.

[0053] For example, in the case where the sheet housing unit **41** is pulled out during the sheet feeding operation, the longer the drive time up to this point, the more the printing paper is dried by the blower fan **35** and the heater **36**, which is sufficiently dried. For this reason, the adhesion between sheets of printing paper is low, and there is a possibility that the alignment of the sheets of

printing paper becomes poor due to the wind force of the blower fan **35**. In this case, the airflow rate control unit **96** reduces the airflow rate of the blower fan **35** as the drive time increases. This prevents the printing paper from being displaced due to the wind force, while the sheet housing unit **41** is pulled out.

[0054] Further, in the case where the sheet housing unit **41** is pulled out during the sheet feeding operation, it is conceivable that the shorter the drive time up to this point, the less the printing paper is dried by the blower fan **35** and the heater **36**, which is insufficiently dried. For this reason, it is conceivable that the adhesion between sheets of printing paper is not reduced and the alignment of the sheets of printing paper is not affected even if the wind force of the blower fan **35** is somewhat strong. In this case, the airflow rate control unit **96** increases the airflow rate of the blower fan **35** as the drive time decreases. This allows the heater **36** to be cooled faster without disturbing the alignment of the sheets of printing paper due to the wind force, while the sheet housing unit **41** is pulled out.

[0055] In another embodiment, the airflow rate control unit **96** may set, in the case where the drive time is equal to or more than the predetermined first threshold value, the set airflow rate of the blower fan **35** to a first airflow rate lower than a predetermined initial set airflow rate. In this case, the airflow rate control unit **96** sets, in the case where the drive time is less than the first threshold value and is equal to or more than the second threshold value, the set airflow rate to a second airflow rate that is higher than the first airflow rate and lower than the initial set airflow rate. Further, the airflow rate control unit **96** sets, in the case where the drive time is less than the second threshold value, the set airflow rate to the initial set airflow rate or a third airflow rate that is higher than the second airflow rate and lower than the initial set airflow rate.

First Processing Example of Sheet Blowing Processing

[0056] An example (first processing example) of the procedure of sheet blowing processing executed by the control unit **90** and a blowing control method according to the present disclosure will be described below with reference to FIG. 5. In FIG. 5, **S11**, **S12**, . . . indicate the numbers (Step numbers) of the processing procedure.

[0057] Note that one or a plurality of Steps included in the sheet blowing processing described below may be omitted as appropriate. Further, the order of execution of Steps in the sheet blowing processing may differ as long as the same operation and effect are achieved. Further, although a case where one processor corresponding to the control unit **90** executes the processing of each Step in the sheet blowing processing will be described below as an example, a plurality of processors may execute the respective Steps in the sheet blowing processing in a distributed manner.

[0058] The sheet blowing processing described below is executed in the case where a printing job for printing an image on printing paper in the sheet housing unit **41** has been input to the image forming apparatus **10** while the sheet housing unit **41** is mounted on the casing **10A**.

[0059] As shown in FIG. 5, in Step **S11**, the control unit **90** acquires type information of the printing paper to be used for the image forming processing. The type information is information indicating the type of printing paper. The possible type of printing paper is, for example, a type identified by various factors such as the basis weight of printing paper, the thickness of printing paper, the presence or absence of coating treatment in printing paper, and the thickness of a coat layer. Specific examples thereof include so-called plain paper that has not been subjected to coating treatment and so-called coated paper that has been subjected to coating treatment.

[0060] For example, sheet-of-paper information including the type information of printing paper housed in the sheet housing unit **41** is stored in the storage unit **94** of the control unit **90**, the control unit **90** acquires the type information of printing paper on which an image is to be formed from the storage unit **94** and determines the type of printing paper in the sheet housing unit **41**.

[0061] In the next Step **S12**, the control unit **90** determines, on the basis of the acquired type information, whether or not warm air needs to be blown onto the printing paper in the sheet housing unit **41**. Specifically, the control unit **90** determines, in the case where the printing paper

housed in the sheet housing unit **41** is coated paper, that warm air needs to be blown and determines, in the case where the printing paper is plain paper, that warm air does not need to be blown. In the case where it is determined that warm air does not need to be blown, the series of processing ends.

[0062] In the case where it is determined in Step **S12** that warm air needs to be blown, the control unit **90** drives the heater **36** to start controlling the heating by the heater **36** (**S13**). Further, the control unit **90** drives the blower fan **35** to start blowing by the blower fan **35**. As a result, warm air is blown onto the printing paper housed in the sheet housing unit **41** (**S14**).

[0063] In Step **S15**, the control unit **90** determines whether or not the sheet housing unit **41** has been pulled out of the casing **10A**. In the case where the sheet housing unit **41** has remained mounted on the casing **10A**, the series of processing ends under the condition that all paper feeding corresponding to the printing job has been completed after that (**S16**).

[0064] Meanwhile, in the case where it is determined in Step **S15** that the sheet housing unit **41** has been pulled out of the casing **10A**, the control unit **90** stops the heater **36** (**S17**) and reduces the airflow rate of the blower fan **35** (**S18**). Note that the degree of reduction in airflow rate can be the airflow rate obtained by multiplying the initial set airflow rate by a predetermined reduction rate (e.g., 50%).

[0065] After that, the control unit **90** determines whether or not the elapsed time after the heater **36** is stopped has elapsed the set time (**S19**), and stops the blower fan **35** in the case where the elapsed time has elapsed the set time (**S20**).

[0066] As described above, in the case where the sheet housing unit **41** is pulled out of the casing **10A** during the sheet feeding operation, the blower fan **35** is continuously driven and the heater **36** is stopped. Therefore, it is possible to prevent the heater **36** from being abnormally overheated and the sheets of printing paper in the sheet housing unit **41** from coming into close contact with each other.

[0067] Note that although the airflow rate of the blower fan **35** is reduced after the heater **36** is stopped in the above-mentioned first processing example, for example, the blower fan **35** may continue to blow at the initial set airflow rate without reducing the airflow rate of the blower fan **35**.

Second Processing Example of Sheet Blowing Processing

[0068] An example (second processing example) of the procedure of sheet blowing processing executed by the control unit **90** will be described below with reference to FIG. **6**. Note that the procedure of the same processing content as that of the above-mentioned first processing example is indicated by the same Step number, and detailed description thereof is omitted.

[0069] As shown in FIG. **6**, when it is determined in Step **S15** that the sheet housing unit **41** has been pulled out, the control unit **90** acquires the drive time during which the heater **36** is driven from the time when driving of the heater **36** started to the time when the sheet housing unit **41** was pulled out of the casing **10A** during the sheet feeding operation (**S151**). The control unit **90** counts, as the drive time, the time from the time point at which driving of the heater **36** started after the input of the printing job to the time point at which the sheet housing unit **41** is pulled out.

[0070] In the next Step **S152**, the control unit **90** changes the set time in accordance with the drive time. That is, the control unit **90** sets the set time to a set value according to the drive time. After that, the above-mentioned processing of Step **S17** and subsequent Steps is executed.

[0071] Even in such a second processing example, it is possible to prevent the heater **36** from being abnormally overheated and the sheets of printing paper in the sheet housing unit **41** from coming into close contact with each other.

Third Processing Example of Sheet Blowing Processing

[0072] An example (third processing example) of the procedure of sheet blowing processing executed by the control unit **90** will be described below with reference to FIG. **7**. Note that the procedure of the same processing content as those of the above-mentioned first processing example

and second processing example is indicated by the same Step number, and detailed description thereof is omitted.

[0073] As shown in FIG. 7, when it is determined in Step S15 that the sheet housing unit 41 has been pulled out, the control unit 90 acquires the drive time.

[0074] In the next Step S152, the control unit 90 determines whether or not the drive time is equal to or more than the first threshold value. The processing of the control unit 90 proceeds to Step S153 in the case where the drive time is equal to or more than the first threshold value and proceeds to Step S155 in the case where the drive time is less than the first threshold value.

[0075] In Step S153, the control unit 90 sets the set time to be used in Step S19 to a first time longer than a predetermined initial set time. Further, in Step S154, the control unit 90 sets the set airflow rate of the blower fan 35 to the first airflow rate lower than the predetermined initial set airflow rate. After that, the above-mentioned processing of Step S17 and subsequent Steps is executed.

[0076] In Step S155, the control unit 90 determines whether or not the drive time is less than the first threshold value and equal to or more than the second threshold value. The processing of the control unit 90 proceeds to Step S156 in the case where the drive time is less than the first threshold value and equal to or more than the second threshold value and proceeds to Step S158 in the case where the drive time is less than the second threshold value.

[0077] In Step S156, the control unit 90 maintains the set time to be used in Step S19 at the initial set time. Further, in Step S157, the control unit 90 sets the set airflow rate of the blower fan 35 to the second airflow rate that is higher than the first airflow rate and lower than the initial set airflow rate. After that, the above-mentioned processing of Step S17 and subsequent Steps is executed.

[0078] Further, in Step S158, the control unit 90 sets the set time to the second time shorter than the initial set time. Further, the control unit 90 sets the set airflow rate to the initial set airflow rate or the third airflow rate that is higher than the second airflow rate and lower than the initial set airflow rate. After that the above-mentioned processing of Step S17 and subsequent Steps is executed.

[0079] Even in such a third processing example, it is possible to prevent the heater 36 from being abnormally overheated and the sheets of printing paper in the sheet housing unit 41 from coming into close contact with each other.

Notes of Invention

[0080] The outline of the invention extracted from the above-mentioned embodiments will be given below. Note that the configurations and processing functions described in the following notes can be arbitrarily selected and combined.

<Note 1>

[0081] A sheet feeding device, including: [0082] a sheet housing unit that is mounted on a body of an apparatus such that the sheet housing unit can be pulled out of the body of the apparatus; [0083] a blowing unit that blows air toward a sheet housed in the sheet housing unit; [0084] a heating unit that heats the air blown by the blowing unit; and [0085] a control unit that controls the blowing unit and the heating unit, [0086] the control unit being configured to [0087] drive the blowing unit and the heating unit during a sheet feeding operation of feeding the sheet from the sheet housing unit, and [0088] continuously drive the blowing unit and stop the heating unit when the sheet housing unit is pulled out of the body of the apparatus during the sheet feeding operation.

<Note 2>

[0089] The sheet feeding device according to Note 2, in which [0090] the control unit is further configured to reduce an airflow rate of the blowing unit from a reference airflow rate to an airflow rate lower than the reference airflow rate when the sheet housing unit is pulled out of the body of the apparatus during the sheet feeding operation.

<Note 3>

[0091] The sheet feeding device according to Note 1 or 2, in which [0092] the control unit is further configured to stop the blowing unit when a predetermined set time has elapsed after the

heating unit is stopped.

<Note 4>

[0093] The sheet feeding device according to Note 3, in which [0094] the control unit is further configured to change the set time in accordance with a drive time of the heating unit until the sheet housing unit is pulled out of the body of the apparatus.

<Note 5>

[0095] The sheet feeding device according to Note 4, in which [0096] the control unit is further configured to increase the set time as the drive time increases and decreases the set time as the drive time decreases.

<Note 6>

[0097] The sheet feeding device according to Note 5, in which [0098] the control unit is further configured to [0099] set, where the drive time is equal to or more than a predetermined first threshold value, the set time to a first time longer than an initial set time, [0100] maintain, where the drive time is less than the first threshold value and equal to or more than a second threshold value that is smaller than the first threshold value, the set time at the initial set time, and [0101] set, where the drive time is less than the second threshold value, the set time to a second time shorter than the initial set time.

<Note 7>

[0102] The sheet feeding device according to any one of Notes 1 to 6, in which [0103] the control unit is further configured to change, in accordance with a drive time of the heating unit until the sheet housing unit is pulled out of the body of the apparatus, an airflow rate of the blowing unit after the heating unit is stopped.

<Note 8>

[0104] The sheet feeding device according to Note 7, in which [0105] the control unit is further configured to [0106] set, where the drive time is equal to or more than a predetermined first threshold value, a set airflow rate of the blowing unit to a first airflow rate lower than an initial set airflow rate, [0107] set, where the drive time is less than the first threshold value and equal to or more than a second threshold value that is smaller than the first threshold value, the set airflow rate to a second airflow rate that is higher than the first airflow rate and lower than the initial set airflow rate, and [0108] set, where the drive time is less than the second threshold value, the set airflow rate to the initial set airflow rate or a third airflow rate that is higher than the second airflow rate and lower than the initial set airflow rate.

<Note 9>

[0109] An image forming apparatus, including: [0110] the sheet feeding device according to any one of Notes 1 to 8; and [0111] an image forming unit that forms an image on the sheet fed by the sheet feeding device.

[0112] It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. A sheet feeding device, comprising: a sheet housing unit that is mounted on a body of an apparatus such that the sheet housing unit can be pulled out of the body of the apparatus; a blowing unit that blows air toward a sheet housed in the sheet housing unit; a heating unit that heats the air blown by the blowing unit; and a control unit that controls the blowing unit and the heating unit, the control unit being configured to drive the blowing unit and the heating unit during a sheet feeding operation of feeding the sheet from the sheet housing unit, and continuously drive the blowing unit and stop the heating unit when the sheet housing unit is pulled out of the body of the

apparatus during the sheet feeding operation.

2. The sheet feeding device according to claim 1, wherein the control unit is further configured to reduce an airflow rate of the blowing unit from a reference airflow rate to an airflow rate lower than the reference airflow rate when the sheet housing unit is pulled out of the body of the apparatus during the sheet feeding operation.

3. The sheet feeding device according to claim 1, wherein the control unit is further configured to stop the blowing unit when a predetermined set time has elapsed after the heating unit is stopped.

4. The sheet feeding device according to claim 3, wherein the control unit is further configured to change the set time in accordance with a drive time of the heating unit until the sheet housing unit is pulled out of the body of the apparatus.

5. The sheet feeding device according to claim 4, wherein the control unit is further configured to increase the set time as the drive time increases and decreases the set time as the drive time decreases.

6. The sheet feeding device according to claim 5, wherein the control unit is further configured to set, where the drive time is equal to or more than a predetermined first threshold value, the set time to a first time longer than an initial set time, maintain, where the drive time is less than the first threshold value and equal to or more than a second threshold value that is smaller than the first threshold value, the set time at the initial set time, and set, where the drive time is less than the second threshold value, the set time to a second time shorter than the initial set time.

7. The sheet feeding device according to claim 1, wherein the control unit is further configured to change, in accordance with a drive time of the heating unit until the sheet housing unit is pulled out of the body of the apparatus, an airflow rate of the blowing unit after the heating unit is stopped.

8. The sheet feeding device according to claim 7, wherein the control unit is further configured to set, where the drive time is equal to or more than a predetermined first threshold value, a set airflow rate of the blowing unit to a first airflow rate lower than an initial set airflow rate, set, where the drive time is less than the first threshold value and equal to or more than a second threshold value that is smaller than the first threshold value, the set airflow rate to a second airflow rate that is higher than the first airflow rate and lower than the initial set airflow rate, and set, where the drive time is less than the second threshold value, the set airflow rate to the initial set airflow rate or a third airflow rate that is higher than the second airflow rate and lower than the initial set airflow rate.

9. An image forming apparatus, comprising: the sheet feeding device according to claim 1; and an image forming unit that forms an image on the sheet fed by the sheet feeding device.
